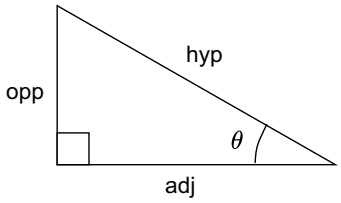
Trigonometry

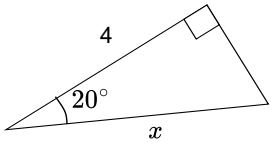
Chapter 9 · Original slide 4

RECAP: Right-Angled Trigonometry

4

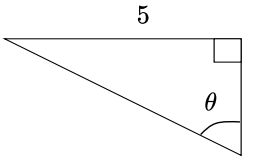


You are probably familiar with the formula: $\sin(\theta) = \frac{opp}{hyp}$
But what is the *conceptual* definition of $\sin$?
sin is a function which inputs an angle and gives the



Find x.

cc
x



Find θ.

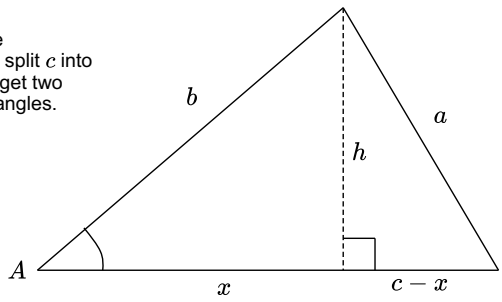
tan
θ =

Chapter 9 · Original slide 5

Proof of Cosine Rule

5

We want to use Pythagoras, so split c into two so that we get two right-angled triangles.



By Pythagoras: $h^2 + x^2 = b^2$ and $h^2 + (c-x)^2 = a^2$
Subtracting to eliminate h:
$x^2 - (c-x)^2 = b^2 - a^2$
$a^2 = b^2 + c^2 - 2cx$
But $x = b\cos A$ ∴
$$a^2 = b^2 + c^2 - 2bc\cos A$$

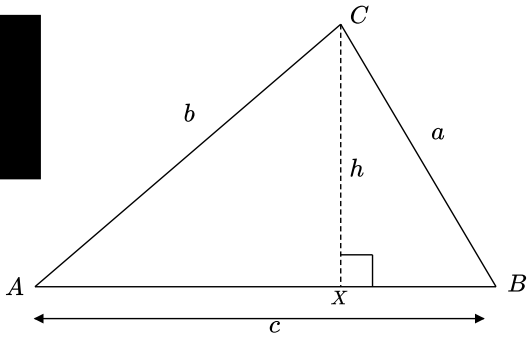
Trigonometry

Chapter 9 · Original slide 6

Proof of Sine Rule

6

The idea is that we can use the common length of ΔACX and $\angle XBC$, i.e. h, to connect the two triangles, and therefore connect their angles/length.



Using basic trigonometry:

$h = a\sin B$ and $h = b\sin A$

$\therefore a\sin B = b\sin A$

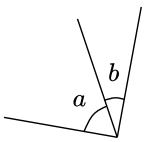
$\therefore \frac{a}{\sin A} = \frac{b}{\sin B}$

Chapter 9 · Original slide 7

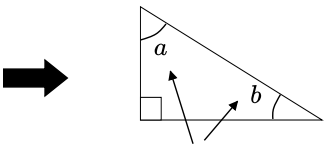
Just for your interest…

7

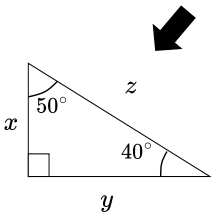
Why does “cosine” contains the word “sine”?



Complementary Angles add to 90°



Therefore these angles are complementary.



$\cos(50) =$ ____

$\sin(40) =$ ____

Trigonometry

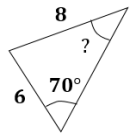
Chapter 9 · Original slide 8

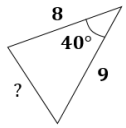
OVERVIEW: Finding missing sides and angles

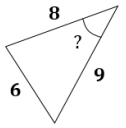
8

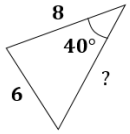
When triangles are not right-angled, we can no longer use simple trigonometric ratios, and must use the cosine and sine rules.

You have	You want
#1: Two angle-side opposite pairs	Missing angle or side in one pair
#2 Two sides known and a missing side opposite a known angle	Remaining side
#3 All three sides	An angle
#4 Two sides known and a missing side not opposite known angle	Remaining side







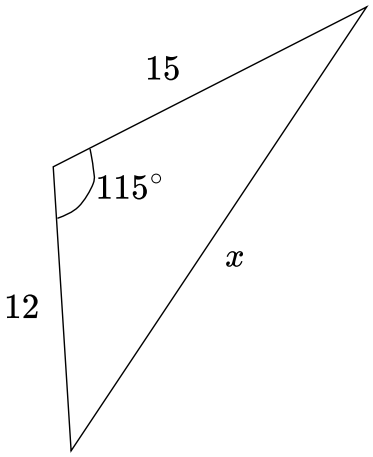


Chapter 9 · Original slide 9

Cosine Rule

9

We use the **cosine rule** whenever we have **three sides** (and an angle) involved.



Proof at end of PowerPoint.

Cosine Rule:

$$a^2 = b^2 + c^2 - 2bc\cos A$$

Trigonometry

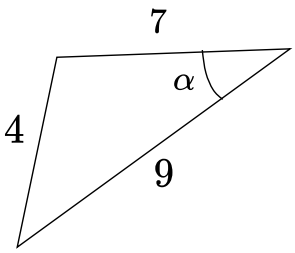
Chapter 9 · Original slide 10

Dealing with Missing Angles

10

You have	You want	Use
All three sides	An angle	Cosine rule

$$\mathbf{a^2=b^2+c^2-2bccosA}$$



Label sides then substitute into formula.

Simplify each bit of formula.

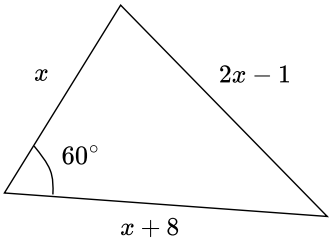
Rearrange "swapping" thing you're subtracting and result

Chapter 9 · Original slide 11

Harder Ones

11

Determine the value of x.



$(2$
$4x$
$3x$
$x^$
$(x$
$\therefore$

[From textbook] Coastguard station B is 8 km, on a bearing of 060° A C

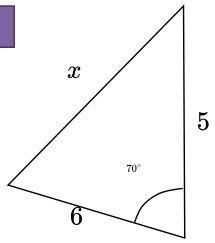
Trigonometry

Chapter 9 · Original slide 12

Test Your Understanding

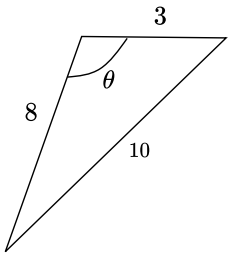
12

1



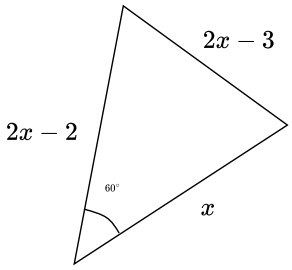
$x =$

2



$\theta =$

3



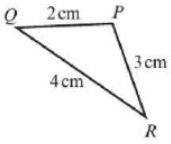
$x =$

Chapter 9 · Original slide 13

Exercise 9A – Y1 pages 177-179

13

8 Show that $\cos P = -\frac{1}{4}$.



11 The lengths of the sides of a triangle are in the ratio $2 : 3 : 4$.
Calculate the size of the largest angle.

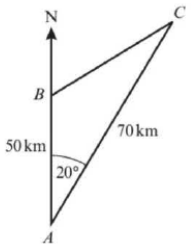
Trigonometry

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Exercise 9A – Y1 pages 177-179

14

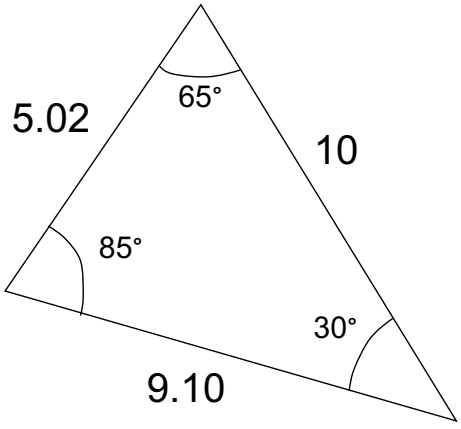
17 The diagram shows three cargo ships, A, B and C, which are in the same horizontal plane. Ship B is 50 km due north of ship A and ship C is 70 km from ship A. The bearing of C from A is $020°$.
a Calculate the distance between ships B and C, in kilometres to 3 s.f. (3 marks)
b Calculate the bearing of ship C from ship B. (4 marks)



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The Sine Rule

16



For this triangle, try calculating each side divided by the sin of its opposite angle. What do you notice in all three cases?

You have	You want	Use
#1. Two angle-side opposite pairs	Missing angle or side in one pair	Sine rule

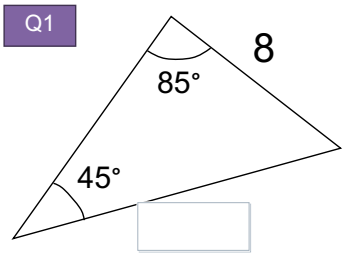
Trigonometry

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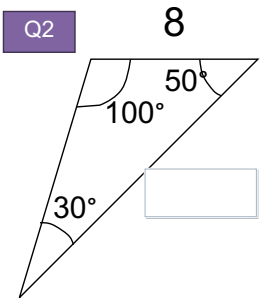
Examples

17

Q1



Q2



You have	You want	Use
#1. Two angle-side opposite pairs	Missing angle or side in one pair	Sine rule

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Examples

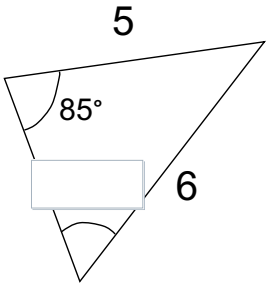
18

When you have a missing angle, it's better to reciprocate to get:

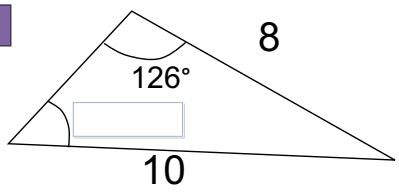
$$\frac{\sin A}{a} = \frac{\sin B}{b}$$

i.e. in general put the missing value in the numerator.

Q3



Q4



Trigonometry

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Exercise 9B – Y1 pages 181-183

19

8 Town B is 6 km, on a bearing of 020°, from town A. Town C is located on a bearing of 055° from town A and on a bearing of 120° from town B. Work out the distance of town C from:
a town A\qquad b town B

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Exercise 9B – Y1 pages 181-183

20

11 In $\triangle ABC$,
$AB = x$ cm, $BC = (4 - x)$ cm, $\angle BAC = y$ and
$\angle BCA = 30^\circ$.
Given that $\sin y = \frac{1}{\sqrt{2}}$, show that

$$x = 4(\sqrt{2} - 1)$$

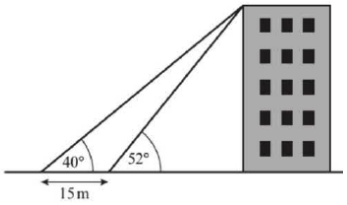
Trigonometry

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Exercise 9B – Y1 pages 181-183

21

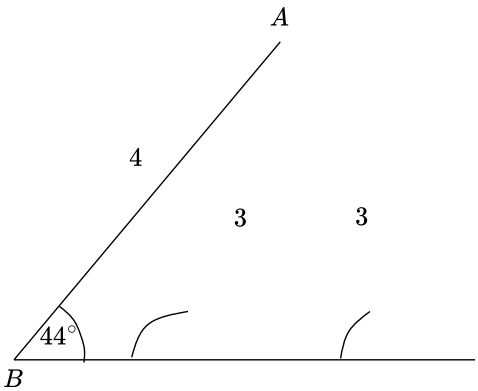
12 A surveyor wants to determine the height of a building. She measures the angle of elevation of the top of the building at two points 15 m apart on the ground.
a Use this information to determine the height of the building. (4 marks)
b State one assumption made by the surveyor in using this mathematical model. (1 mark)



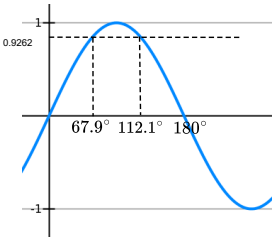
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The ‘Ambiguous Case’

23



Suppose you are told that $AB = 4$, $AC = 3$ and $\angle ABC = 44^\circ$. What are the possible values of $\angle ACB$?



Your calculator will give the acute angle of 67.9° (i.e. C_2). But if we look at a graph of sin, we can see there’s actually a second value for $\sin^{-1}(0.9262)$, corresponding to angle C_1.

The sine rule produces two possible solutions for a missing angle:

$$\sin\theta = \sin(180^\circ - \theta)$$

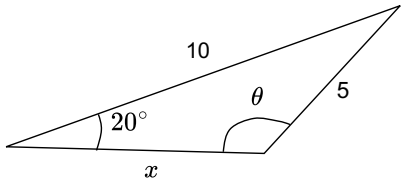
Whether we use the acute or obtuse angle depends on context.

Trigonometry

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Test Your Understanding

24



Given that the angle θ is obtuse, determine θ and hence determine the length of x.

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Exercise 9C – Y1 pages 184-185

25

4 Triangle ABC is such that $AB = 4$ cm, $BC = 6$ cm and $\angle ACB = 36°$. Show that one of the possible values of $\angle ABC$ is $25.8°$ (to 3 s.f.). Using this value, calculate the length of AC.

Trigonometry

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Exercise 9C – Y1 pages 184-185

26

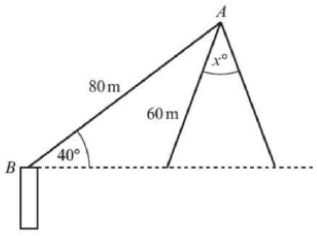
5 Two triangles ABC are such that $AB = 4.5\text{ cm}$, $BC = 6.8\text{ cm}$ and $\angle ACB = 30°$. Work out the value of the largest angle in each of the triangles.

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Exercise 9C – Y1 pages 184-185

27

6 a A crane arm AB of length 80 m is anchored at point B at an angle of $40°$ to the horizontal. A wrecking ball is suspended on a cable of length 60 m from A. Find the angle x through which the wrecking ball rotates as it passes the two points level with the base of the crane arm at B. (6 marks)
b Write down one modelling assumption you have made. (1 mark)

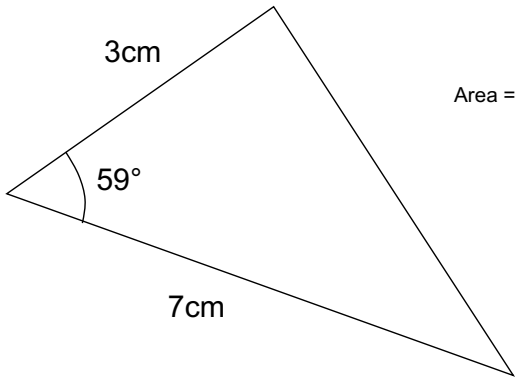


Trigonometry

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Area of Non Right-Angled Triangles

28



Area =

$$\text{Area} = \frac{1}{2}ab\sin(C)$$

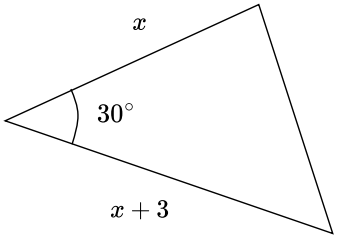
where C is the angle between two sides a and b.

Tip: You shouldn't have to label sides/angles before using the formula. Just remember that the angle is between the two sides.

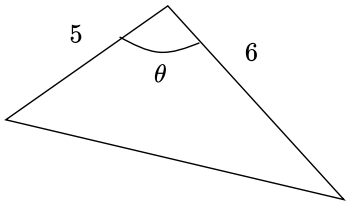
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Test Your Understanding

29



The area of this triangle is 10.
Determine x.



The area of this triangle is also 10.
If θ is obtuse, determine θ.

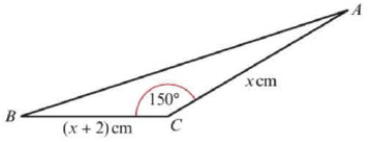
Trigonometry

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Exercise 9D – Y1 pages 186-187

30

4 In triangle ABC, $BC = (x + 2)$ cm, $AC = x$ cm and $\angle BCA = 150^\circ$. Given that the area of the triangle is 5 cm^2, work out the value of x, giving your answer to 3 significant figures.



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Exercise 9D – Y1 pages 186-187

31

5 In $\triangle PQR$, $PQ = (x + 2)$ cm, $PR = (5 - x)$ cm and $\angle QPR = 30^\circ$. The area of the triangle is A cm^2.

a Show that $A = \frac{1}{4}(10 + 3x - x^2)$.

b Use the method of completing the square, or otherwise, to find the maximum value of A, and give the corresponding value of x.

Trigonometry

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Exercise 9D – Y1 pages 186-187

32

6 In $\triangle ABC$, $AB = x$ cm, $AC = (5 + x)$ cm and $\angle BA$
that the area of the triangle is $3\frac{3}{4}$ cm^2
a Show that x satisfies the equation $x^2 + 5x - 15 = 0$.
b Calculate the value of x, giving your answer to 3 significa... figures.

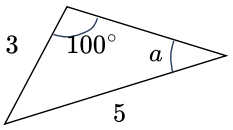
Chapter 9 · Original slide 33

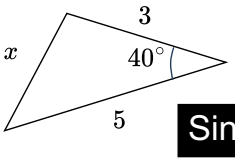
Sin or cosine rule?

33

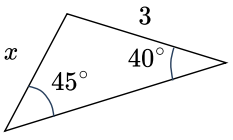
Recall that whenever we have **two "side-angle pairs"** involved, use sine rule. If there's **3 sides** involved, we can use cosine rule. Sine rule is generally easier to use than cosine rule.

Sine Cosine

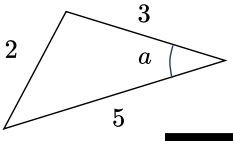




Sine Cosine



Sine Cosine



Sine Cosine

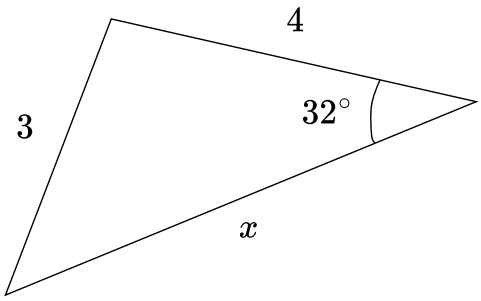
Trigonometry

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Using sine rule twice

34

You have	You want	Use
#4 Two sides known and a missing side not opposite known angle	Remaining side	Sine rule twice



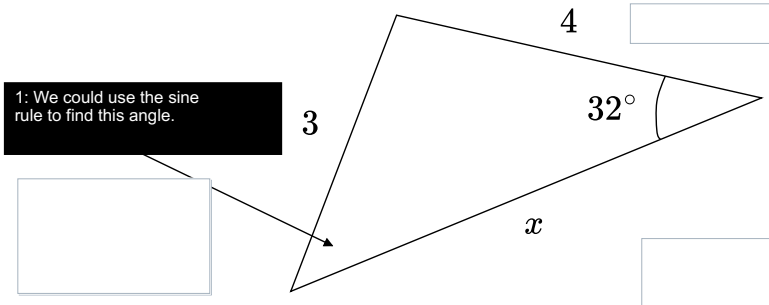
Given there is just one angle involved, you might attempt to use the cosine rule:

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Using sine rule twice

35

You have	You want	Use
#4 Two sides known and a missing side not opposite known angle	Remaining side	Sine rule twice



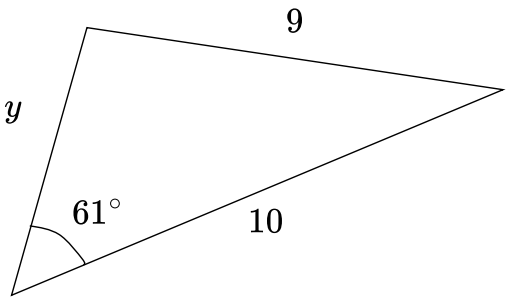
1: We could use the sine rule to find this angle.

Trigonometry

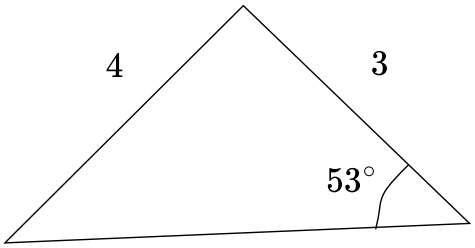
Chapter 9 · Original slide 36

Test Your Understanding

36



$y =$



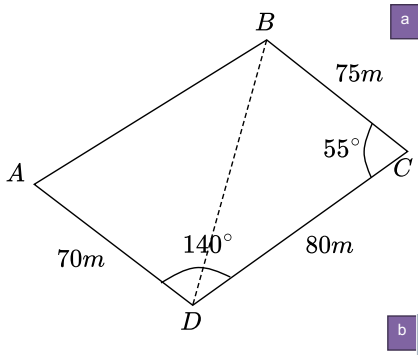
$Area =$

Chapter 9 · Original slide 37

Problem Solving With Sine/Cosine Rule

37

[From Textbook] The diagram shows the locations of four mobile phone masts in a field, $BC = 75m$. $CD = 80m$, angle $BCD = 55°$ and angle $ADC = 140°$.
In order that the masts do not interfere with each other, they must be at least 70m apart.
Given that A is the minimum distance from D, find:
a) The distance A is from B
b) The angle BAD
c) The area enclosed by the four masts.



a

b

c

Trigonometry

Chapter 9 · Original slide 38

Exercise 9E – Y1 page 189-191

38

12 The perimeter of $\triangle ABC = 15\ \text{cm}$. Given that $AB = 7\ \text{cm}$ and $\angle BAC = 60°$, find the lengths of AC and BC and the area of the triangle.

Chapter 9 · Original slide 39

Exercise 9E – Y1 page 189-191

39

13 In the triangle ABC, $AB = 14\ \text{cm}$, $BC = 12\ \text{cm}$ and $CA = 15\ \text{cm}$.
a Find the size of angle C, giving your answer to 3 s.f.
b Find the area of triangle ABC, giving your answer in cm^2 to 3 s.f.

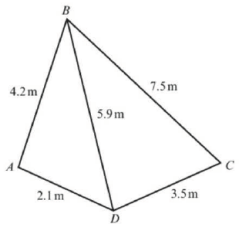
Trigonometry

Chapter 9 · Original slide 40

Exercise 9E – Y1 page 189-191

40

14 A flower bed is in the shape of a quadrilateral as shown in the diagram.
a Find the sizes of angles DAB and BCD. \hfill(4 ma...
b Find the total area of the flower bed. \hfill(3 ma...
c Find the length of the diagonal AC.

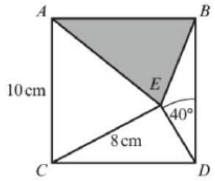


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Exercise 9E – Y1 page 189-191

41

15 $ABCD$ is a square. Angle CED is obtuse.
Find the area of the shaded triangle. (7 marks)



Trigonometry

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Sin Graph

44

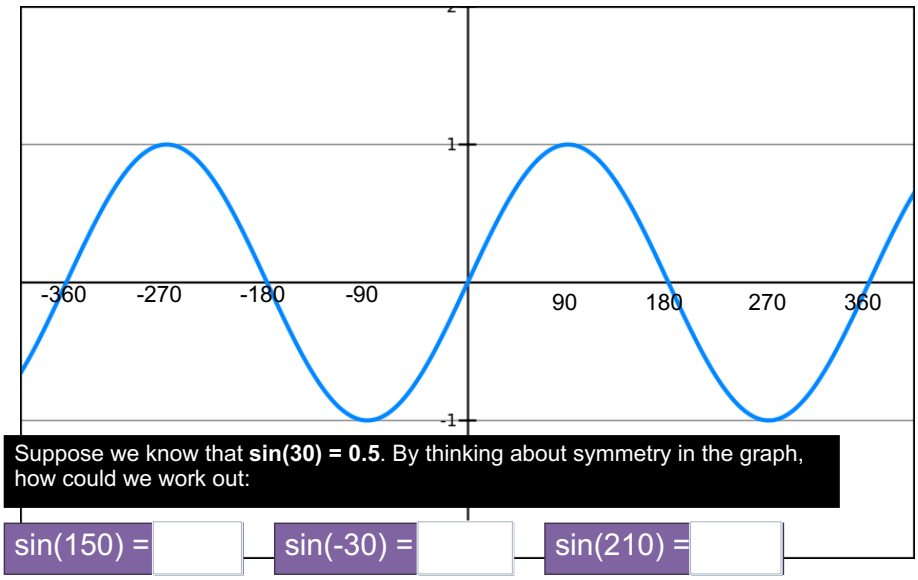
What does it look like?

Chapter 9 · Original slide 45

Sin Graph

45

What do the following graphs look like?



Suppose we know that **sin(30) = 0.5**. By thinking about symmetry in the graph, how could we work out:

sin(150) = sin(-30) = sin(210) =

Trigonometry

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Cos Graph

46

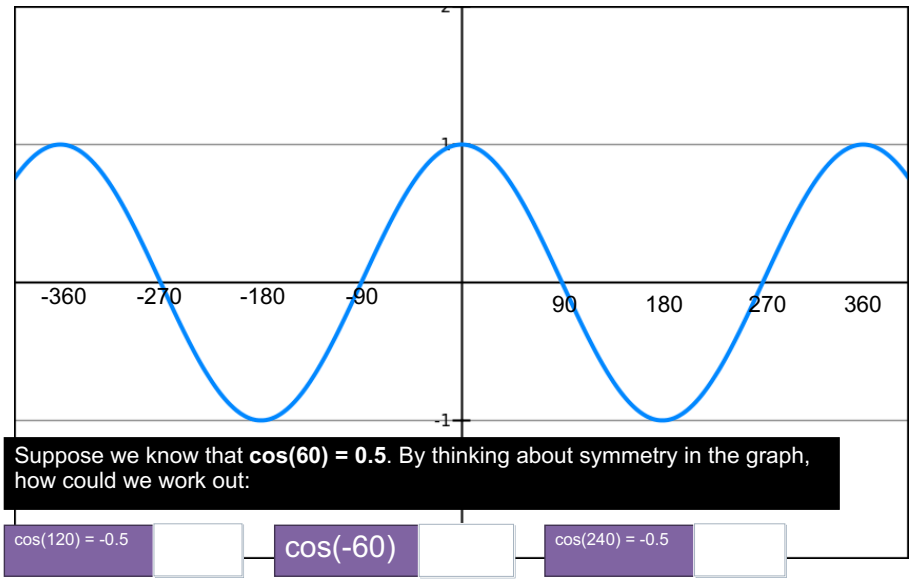
What do the following graphs look like?

Chapter 9 · Original slide 47

Cos Graph

47

What does it look like?



Suppose we know that **cos(60) = 0.5**. By thinking about symmetry in the graph, how could we work out:

cos(120) = -0.5

cos(-60)

cos(240) = -0.5

Trigonometry

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Tan Graph

48

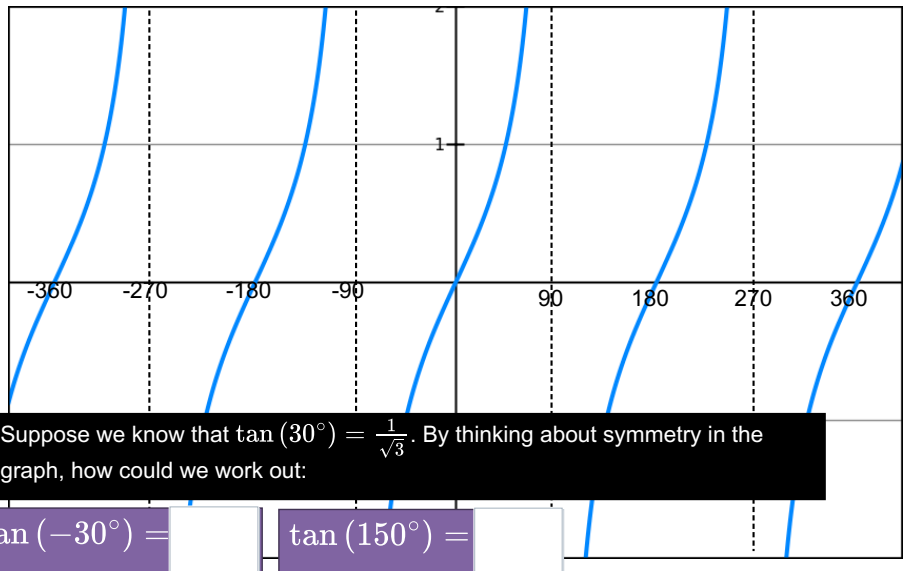
What does it look like?

Chapter 9 · Original slide 49

Tan Graph

49

What does it look like?



Suppose we know that $\tan(30^\circ) = \frac{1}{\sqrt{3}}$. By thinking about symmetry in the graph, how could we work out:

$\tan(-30^\circ) =$ $\tan(150^\circ) =$

Trigonometry

Chapter 9 · Original slide 50

Transforming Trigonometric Graphs

50

There is no new theory here: just use your knowledge of transforming graphs, i.e. whether the transformation occurs ‘inside’ the function (i.e. input modified) or ‘outside’ the function (i.e. output modified).

Sketch $y = 4\sin x,\ 0 \leq x \leq 360^\circ$

Sketch $y = \cos(x + 45^\circ),\ 0 \leq x \leq 360^\circ$

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Transforming Trigonometric Graphs

51

Sketch $y = -\tan x,\ 0 \leq x \leq 360^\circ$

Sketch $y = \sin\left(\frac{x}{2}\right),\ 0 \leq x \leq 360^\circ$

Trigonometry

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Exercise 9F – Y1 Page 194

52

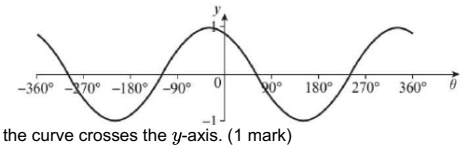
1 Sketch the graph of $y = \cos\theta$ in the interval $-180^\circ \leq \theta \leq 180^\circ$.\par2 Sketch the graph of $y = \tan\theta$ in the interval $-180^\circ \leq \theta \leq 180^\circ$.\par3 Sketch the graph of $y = \sin\theta$ in the interval $-90^\circ \leq \theta \leq 270^\circ$.

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Exercise 9G – Y1 Pages 197-198

53

7 The graph shows the curve
$y = \cos(x + 30^\circ), -360^\circ \leq x \leq 360^\circ$.
a Write down the coordinates of the points where the curve crosses the x-axis. (2 marks)



b Find the coordinates of the point where the curve crosses the y-axis. (1 mark)

b $\cos 30^\circ = \frac{\sqrt{3}}{2}$; $\left(0, \frac{\sqrt{3}}{2}\right)$

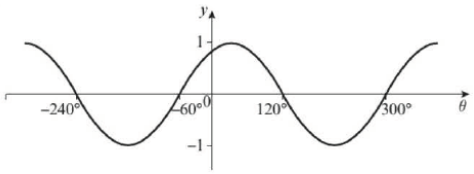
Trigonometry

Chapter 9 · Original slide 54

Exercise 9G – Y1 Pages 197-198

54

8 The graph shows the curve with equation $y = \sin(x + k), -360^\circ \leq x \leq 360^\circ$, where k is a constant.
a Find one possible value for k. (2 marks)
b Is there more than one possible answer to part a? Give a reason for your answer. (2 marks)



Chapter 9 · Original slide 55

Exercise 9G – Y1 Pages 197-198

55

9 The variation in the depth of water in a rock pool can be modelled using the function $y = \sin(30t)^\circ$, where t is the time in hours and $0 \leq t \leq 6$.
a Sketch the function for the given interval.
b If $t = 0$ represents midday, during what times will the rock pool be at least half full?

Trigonometry

Chapter 10 · Original slide 3

sin/cos/tan of $30°, 45°, 60°$

3

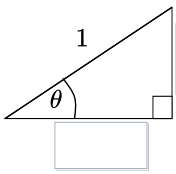
You will frequently encounter angles of $30°, 60°, 45°$ in geometric problems. Why?

Chapter 10 · Original slide 4

The Unit Circle and Trigonometry

4

For values of θ in the range $0 < \theta < 90°$, you know that $\sin\theta$ and $\cos\theta$ are lengths on a right-angled triangle:



Trigonometry

Chapter 10 · Original slide 5

Mini-Exercise

5

Use the unit circle to determine each value in the table, **using either "0", "+ve", "-ve", "1", "-1" or "undefined"**.
Recall that the point on the unit circle has coordinate $(\cos\theta, \sin\theta)$ and OP has gradient $\tan\theta$.

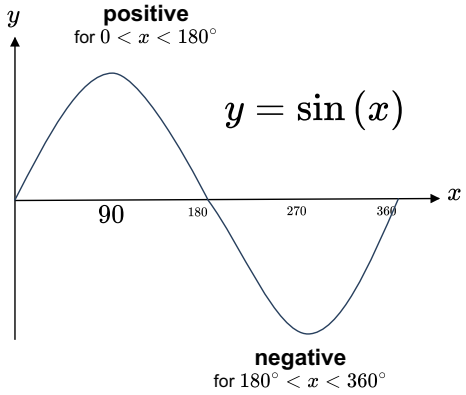
	$\cos\theta$	$\sin\theta$	$\tan\theta$		$\cos\theta$	$\sin\theta$	$\tan\theta$
$\theta = 0$ 	1	0	0	$\theta = 180^\circ$			
$0 < \theta < 90^\circ$ 				$180^\circ < \theta < 270^\circ$			
$\theta = 90^\circ$				$\theta = 270^\circ$			
$90^\circ < \theta < 180^\circ$				$270^\circ < \theta < 360^\circ$			

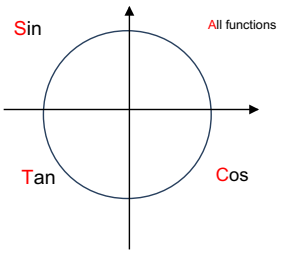
Chapter 10 · Original slide 6

The Unit Circle and Trigonometry

6

The unit circles explains the behaviour of these trigonometric graphs beyond 90°.
However, the easiest way to remember whether $\sin(x), \cos(x), \tan(x)$ are positive or negative is to just do a **very quick sketch (preferably mentally!)** of the corresponding graph.





Note: The textbook uses something called '*CAST diagrams*', which is an acronym for which quadrants the **C**osine, **A**ll, **S**ine and **T**angent are positive

Trigonometry

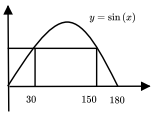
Chapter 10 · Original slide 7

A Few Trigonometric Angle Laws

7

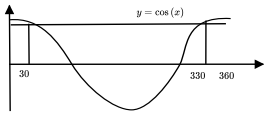
The following are all easily derivable using a quick sketch of a trigonometric graph, and are merely a convenience so you don't always have to draw out a graph every time.
You are highly encouraged to **memorise these** so that you can do exam questions faster.

1. $\sin(x) = \sin(180^\circ - x)$
   e.g. $\sin(150^\circ) = \sin(30^\circ)$

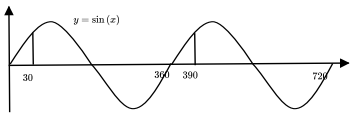


We saw this in the previous chapter when covering the 'ambiguous case' when using the sine rule.

2. $\cos(x) = \cos(360^\circ - x)$
   e.g. $\cos(330^\circ) = \cos(30^\circ)$



3. $\sin$ and $\cos$ repeat every 360° but $\tan$ every 180°
   e.g. $\sin(390^\circ) = \sin(30^\circ)$



4. $\sin(x) = \cos(90^\circ - x)$
   e.g. $\sin(50^\circ) = \cos(40^\circ)$

Remember from the previous chapter that "cosine" by definition is the sine of the "complementary" angle. This was/is never covered in the textbook but caught everyone by surprise when it came up in a C3 exam.

Chapter 10 · Original slide 8

Examples

8

Without a calculator, work out the value of each below.

$\tan(225^\circ) =$
$\tan(210^\circ) =$
$\sin(150^\circ) =$
$\cos(300^\circ) =$
$\sin(-45^\circ) =$
$\cos(750^\circ) =$
$\cos(120^\circ) =$

$-\frac{1}{\sqrt{2}}$
$\frac{\sqrt{3}}{2}$

Use the 'laws' where you can, but otherwise just draw out a quick sketch of the graph.

- $\sin(x) = \sin(180 - x)$
- $\cos(x) = \cos(360 - x)$
- $\sin, \cos$ repeat every 360° but $\tan$ every 180°

Trigonometry

Chapter 10 · Original slide 9

Test Your Understanding

9

Without a calculator, work out the value of each below.

$\cos(315^\circ) =$ $\frac{1}{\sqrt{2}}$

$\sin(420^\circ) =$

$\tan(-120^\circ) =$

$\tan(-45^\circ) =$ -1

$\sin(135^\circ) =$

- $\sin(x) = \sin(180 - x)$
- $\cos(x) = \cos(360 - x)$
- $\sin, \cos$ repeat every 360° but $\tan$ every 180°

Chapter 10 · Original slide 10

Exercise 10B – Y1 Page 209

10

1 Express the following as trigonometric ratios of either 30°, 45° or 60°, and hence find their exact values.\quad
a $\sin 135^\circ$\quad**b** $\sin(-60^\circ)$\quad**c** $\sin 330^\circ$\quad**d** $\sin 420^\circ$\quad**e** $\sin(-300^\circ)$\quad**f** $\cos 120^\circ$\quad**g** $\cos 300^\circ$\quad**h** $\cos 225^\circ$\quad**i** $\cos(-210^\circ)$\quad**j** $\cos 495^\circ$\quad**k** $\tan 135^\circ$\quad**l** $\tan(-225^\circ)$\quad**m** $\tan 210^\circ$\quad**n** $\tan 300^\circ$\quad**o** $\tan(-120^\circ)$.

Trigonometry

Chapter 10 · Original slide 11

Exercise 10A/B – Y1 Page 207, 209

11

Challenge

The diagram shows an isosceles right-angled triangle ABC.

$AE = DE = 1$ unit. Angle $ACD = 30°$.

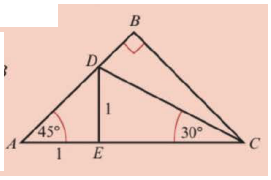
a Calculate the exact lengths of

i CE \quad **ii** DC \quad **iii** BC \quad **iv** DB

b State the size of angle BCD.

c Hence find exact values for

i $\sin 15°$ \quad **ii** $\cos 15°$

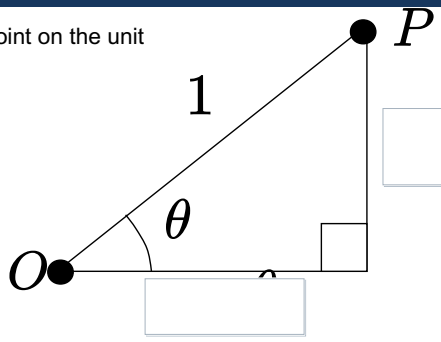


Chapter 10 · Original slide 12

Trigonometric Identities

12

Returning to our point on the unit circle…



1 **Then** $\tan\theta =$

2 Pythagoras gives you...

Trigonometry

Chapter 10 · Original slide 13

Application of identities #1: Proofs

13

Prove that $1 - \tan\theta \sin\theta \cos\theta \equiv \cos^2\theta$

$LHS =$

$= 1 -$

$= 1 - \sin$

$= \cos^2\theta$

$$\tan x = \frac{\sin(x)}{\cos(x)}$$
$$\sin^2 x + \cos^2 x = 1$$

Recall that $\equiv$ means 'equivalent to', and just means the LHS is **always** equal to the RHS for all values of θ.
From Chapter 7 ('Proofs') we saw that usually the best method is to manipulate one side (e.g. LHS) until we get to the other (RHS).

Chapter 10 · Original slide 14

More Examples

14

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Q16

Prove that $\tan\theta + \frac{1}{\tan\theta} \equiv \frac{1}{\sin\theta\cos\theta}$

$LHS \equiv$

$\frac{\cos^2\theta}{\sin\theta\cos\theta}$

$\equiv$

Tip #2: In any addition/subtraction involving at least one fraction (with trig functions), always combine algebraically into one.

Simplify $5 - 5\sin^2\theta$

Trigonometry

Chapter 10 · Original slide 15

Test Your Understanding

15

Prove that $\frac{\tan x \cos x}{\sqrt{1-\cos^2 x}} \equiv 1$

Prove that $\frac{\cos^4\theta - \sin^4\theta}{\cos^2\theta} \equiv 1 - \tan^2\theta$

L
$\equiv$
$\equiv$

AQA IGCSE Further Maths Worksheet

Prove that $\tan^2\theta \equiv \frac{1}{\cos^2\theta} - 1$

Chapter 10 · Original slide 16

Exercise 10C – Y1 Page 211-212

16

10 In each of the following, eliminate θ to give an equation relating x and y:

a $x = \sin\theta, y = \cos\theta$ **b** $x = \sin\theta, y = 2\cos\theta$

c $x = \sin\theta, y = \cos^2\theta$ **d** $x = \sin\theta, y = \tan\theta$

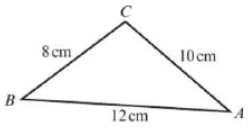
Trigonometry

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Exercise 10C – Y1 Page 211-212

17

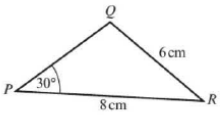
11 The diagram shows the triangle ABC with $AB = 12\,\text{cm}$, $BC = 8\,\text{cm}$ and $AC = 10\,\text{cm}$.



a Show that $\cos B = \frac{9}{16}$. \hfill (3 ma

b Hence find the exact value of $\sin B$.

12 The diagram shows triangle PQR with $PR = 8\,\text{cm}$, $QR = 6\,\text{cm}$ and angle $QPR = 30°$.



a Show that $\sin Q = \frac{2}{3}$.

b Given that Q is obtuse, find the exact value of $\cos Q$.

Chapter 10 · Original slide 19

Solving Trigonometric Equations

19

Remember those trigonometric angle laws (on the right) earlier this chapter? They're about to become very useful!

Reminder of 'trig laws':
- $\sin(x) = \sin(180 - x)$
- $\cos(x) = \cos(360 - x)$
- sin, cos repeat every $360°$ but tan every $180°$

Solve $\sin\theta = \frac{1}{2}$ in the interval $0 \leq \theta \leq 360°$.

Solve $5\tan\theta = 10$ in the interval $-180° \leq \theta < 180°$

Trigonometry

Chapter 10 · Original slide 20

Slightly Harder Ones…

20

Solve $\sin\theta = -\frac{1}{2}$ in the interval $0 \leq \theta \leq 360°$.

Solve $\sin\theta = \sqrt{3}\cos\theta$ in the interval $0 \leq \theta \leq 360°$.

Hint: The problem here is that we have two different trig functions. Is there anything we can divide both sides by so we only have one trig function?

Chapter 10 · Original slide 21

Test Your Understanding

21

Solve $2\cos\theta = \sqrt{3}$ in the interval $0 \leq \theta \leq 360°$.

Solve $\sqrt{3}\sin\theta = \cos\theta$ in the interval $-180° \leq \theta \leq 180°$.

Trigonometry

Chapter 10 · Original slide 22

Exercise 10D – Y1 215-216

22

7 A teacher asks two students to solve the equation $2\cos x = 3\sin x$ for $-180° \leq x \leq 180°$. The attempts are shown:
Student A: $\tan x = \frac{3}{2}$, $x = 56.3°$ or $x = -123.7°$.
Student B: $4\cos^2 x = 9\sin^2 x$, $4(1 - \sin^2 x) = 9\sin^2 x$, $4 = 13\sin^2 x$, $\sin x = \pm\sqrt{\frac{4}{13}}$, $x = \pm 33.7°$ or $x = \pm 146.3°$.
a Identify the mistake made by Student A. (1 mark)
b Identify the mistake made by Student B and explain the effect it has on their solution. (2 marks)
c Write down the correct answers to the question. (1 mark)

9 Find all the values of θ, to 1 decimal place, in the interval $0 < \theta < 360°$ for which $\tan^2\theta = 9$

Chapter 10 · Original slide 23

Exercise 10D – Y1 215-216

23

10 a Show that $4\sin^2 x - 3\cos^2 x = 2$ can be written as $7\sin^2 x = 5$.
b Hence solve, for $0 \leq x \leq 360°$, the equation $4\sin^2 x - 3\cos^2 x = 2$. Give your answers to 1 decimal place.

11 a Show that the equation $2\sin^2 x + 5\cos^2 x = 1$ can be written as $3\sin^2 x = 4$.
b Use your result in part a to explain why the equation $2\sin^2 x + 5\cos^2 x = 1$ has no solutions.

Trigonometry

Chapter 10 · Original slide 24

Harder Equations

24

Harder questions replace the angle θ with a linear expression.

Solve $\cos 3x = -\frac{1}{2}$ in the interval $0 \leq x \leq 360^\circ$.

STEP 1: Adjust the range of values for θ to match the expression inside the cos.

STEP 2: Immediately after applying an inverse trig function (and BEFORE dividing by 3!), find all solutions up to the end of the interval.

STEP 3: Then do final manipulation to each value.

Chapter 10 · Original slide 25

Further Examples

25

Solve $\sin(2x + 30^\circ) = \frac{1}{\sqrt{2}}$ in the interval $0 \leq x \leq 360^\circ$.

Solve $\sin x = 2\cos x$ in the interval $0 \leq x < 300^\circ$

Trigonometry

Chapter 10 · Original slide 26

Test Your Understanding

26

Edexcel C2 Jan 2013 Q4

Solve, for $0 \leq x < 180^\circ$,
$\cos(3x - 10^\circ) = -0.4$,
giving your answers to 1 decimal place. You should show each step in your working. (7)

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Exercise 10E – Y1 p219

27

4 Solve for $0 \leq x \leq 180^\circ$ the equations:
a $\sin(x + 20^\circ) = \frac{1}{2}$.
b $\cos 2x = -0.8$, giving your answers to 1 decimal place.

5 a Sketch for $0 \leq x \leq 360^\circ$ the graph of $y = \sin(x + 60^\circ)$.
b Write down the exact coordinates of the points where the graph meets the coordinate axes.
c Solve, for $0 \leq x \leq 360^\circ$, the equation $\sin(x + 60^\circ) = 0.55$, giving your answers to 1 decimal place.

Trigonometry

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Exercise 10E – Y1 p219

28

6 a Given that $4\sin x = 3\cos x$, write down the value of $\tan x$.
b Solve, for $0 \leq \theta \leq 360^\circ$, $4\sin 2\theta = 3\cos 2\theta$ giving your answers to 1 decimal place.

7 The equation $\tan kx = -\frac{1}{\sqrt{3}}$, where k is a constant and $k > 0$, has a solution at $x = 60^\circ$.
a Find a possible value of k.
b State, with justification, whether this is the only such possible value of k.

Chapter 10 · Original slide 29

Quadratics in sin/cos/tan

29

We saw that an equation can be 'quadratic in' something, e.g. $x - 2\sqrt{x} + 1 = 0$ is 'quadratic in $\sqrt{x}$', meaning that $\sqrt{x}$ could be replaced with another variable, say y, to produce a quadratic equation $y^2 - 2y + 1 = 0$.

Solve $5\sin^2 x + 3\sin x - 2 = 0$ in the interval $0 \leq x \leq 360^\circ$.

Method 1: Use a substitution.

Method 2: Factorise without substitution.

Trigonometry

Chapter 10 · Original slide 30

More Examples

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Solve $\tan^2\theta = 4$ in the interval $0 \leq x \leq 360^\circ$.

Solve $2\cos^2 x + \sin x = 3\sin^2 x$ in the interval $-180^\circ \leq x \leq 180^\circ$.

Tip: We have an identity involving sin^2 and cos^2, so it makes sense to change the squared one that would match all the others.

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Test Your Understanding

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(a) Show that the equation $5\sin x = 1 + 2\cos^2 x$ can be written in the form $2\sin^2 x + 5\sin x - 3 = 0$. (2)
(b) Solve, for $0 \leq x < 360^\circ$, $2\sin^2 x + 5\sin x - 3 = 0$. (4)

Trigonometry

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Exercise 10F – Y1 P221-222

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5 Find all the solutions, in the interval $0 \leq x \leq 360^\circ$, to the equation $8\sin^2 x + 6\cos x - 9 = 0$ giving each solution to one decimal place. (6 marks)

6 Find, for $0 \leq x \leq 360^\circ$, all the solutions of $\sin^2 x + 1 = \frac{7}{2}\cos^2 x$ giving each solution to one decimal place.

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Exercise 10F – Y1 P221-222

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7 Show that the equation $2\cos^2 x + \cos x - 6 = 0$ has no solutions.

8 a Show that the equation $\cos^2 x = 2 - \sin x$ can be written as $\sin^2 x - \sin x + 1 = 0$. (2 marks)

b Hence show that the equation $\cos^2 x = 2 - \sin x$ has no solutions. (3 marks)

Trigonometry

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9\quad $\tan^2 x - 2\tan x - 4 = 0$.
a Show that $\tan x = p \pm \sqrt{q}$ where p and q are numbers to be found.
b Hence solve the equation $\tan^2 x - 2\tan x - 4 = 0$ in the interval $0 \leq x \leq 540°$.